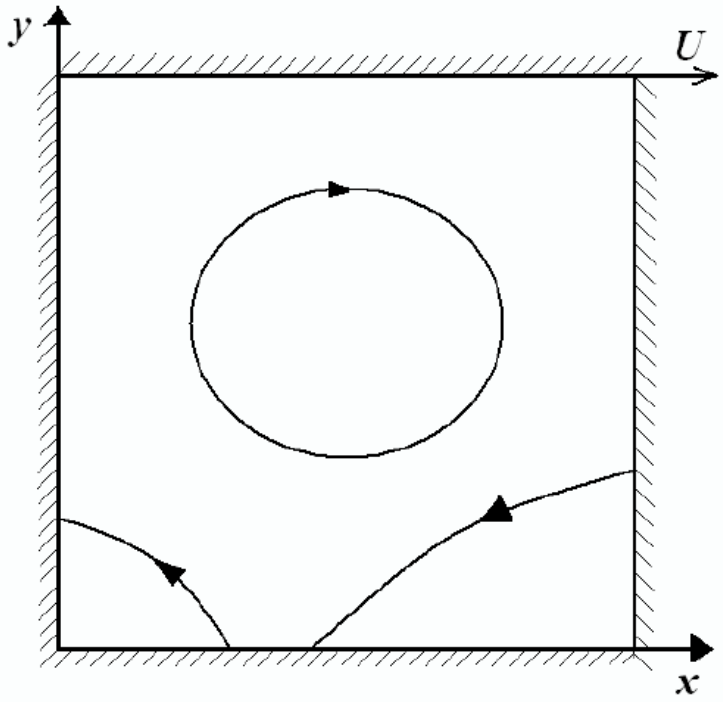
MEE 252B, Final Project

Due Date: **Thursday, December 3, 2026, 9a.m.**

Incompressible flow in a driven cavity

Calculate the incompressible, viscous, rotational flow in a square cavity whose lid is moving to the right at constant velocity U.



The aspect ratio of the cavity is one. The walls are located at $x = 0$, $x = 1$, $y = 0$, $y = 1$. Find steady-state results for Reynolds numbers of 1, 10, and 100 by using the pseudotransient approach. Employ a finite difference approach with second order central differencing in space where possible, based on the primitive variable formulation of the two-dimensional Navier-Stokes equations. Employ the projection method to satisfy the continuity equation. Solve the parabolic

momentum equations by means of the FTCS method, and the elliptic Poisson equation by SOR. Use 8 points in each direction initially, and increase the resolution to up to 64×64 points as needed. Check the accuracy of your results by comparing simulations of different resolutions.

For the project report, document your work in detail. Clearly state the governing equations you use, along with the boundary conditions. Explain the implementation of the equations and boundary conditions in your computer program. Attach a listing of your program to the documentation.

Proceed along the following lines:

1) Formulate the problem in terms of the u, v, p variables. Establish the appropriate boundary conditions.

2) Write and test the individual components of your computer program. Provide information on how you tested the various components of your program, e.g. on testing your SOR procedure with known solutions of the Laplace equation etc.

3) Document your results. Don't list pages of numbers. Rather present your results graphically. E.g., provide contour plots of the vorticity variable. Present $u(y)$ at $x = 0.5$, and $v(x)$ at $y = 0.5$.

4) Compare your results with those given in the paper by J. Kim and P. Moin: Application of a fractional step method to incompressible Navier-Stokes equations. J. Comp. Physics, vol. 59, p. 308 (1985). Other papers that you should compare with are references [16] and [18] given in this paper.

Please turn the project in on time, by email to meiburg@ucsb.edu. If you have questions, see me in my office or email me.